Igor Szuniewicz

Technical Sound Designer | Audio Programmer

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PROFESSIONAL SUMMARY

I design and implement interactive audio for games and VR, and build tools when the workflow needs one. Recent work includes spatial VR audio at XR Labs in Brussels, Wwise and C++ systems in Unreal, a C# beatbox classifier, and released C++/JUCE, Lua/ReaScript and Python tools.

TECHNICAL SKILLS

Game audio: Sound design, audio implementation, spatial audio, interactive mixing, recording, editing, adaptive audio, music composition

Engines and middleware: Unreal Engine 5, Unity 6, Wwise, Steam Audio, MetaSounds, FMOD

Programming and DSP: C++, C#, Python, Lua/ReaScript, Blueprints, JUCE, FFT, onset detection, KNN

Tools and workflow: REAPER, ReaImGui, Perforce, Git, GitHub

EXPERIENCE

Technical Sound Designer | XR Labs Belgium **February 2026 - June 2026**

Brussels, Belgium | Unity | Steam Audio | Perforce | Selected VR and AR training simulations

- Designed and implemented audio for selected Unity VR and AR training simulations.
- Implemented Steam Audio direction, occlusion and indirect paths when walls blocked the direct route.
- Built near and far weapon-shot layers so close detail faded into the distance layer as the listener moved away.

SELECTED PROJECTS

AI Audio Research | Sole Researcher and Prototype Developer **Research 2024 - 2025 | thesis 2026**

Unreal Engine 5 | Python | 68-page thesis

- Built an Unreal client, Python generation server, game-state pre-generation, cache and procedural fallback to test AI-generated sound from an in-game event to playback.
- Timed one procedural baseline and four AI configurations from an in-game event to the first audible sample. At the 95th percentile, DSP took 25 ms and the AI methods 1.9-3.5 seconds against a 150 ms upper budget.

Shadow Frames | Sole Audio Designer **2025**

Wwise | Unreal Engine 5 | C++ | REAPER | Four-month team project

- Created the cues that announce and locate ten environmental changes, then implemented and mixed all game audio in Wwise and Unreal.
- Built spatial clues, surface-aware footsteps, designer-facing settings and game-state mix rules.

Transientica | Solo Developer **2025 - Present**

Unity 6 | C# | KNN | FFT | In development

- Built a rhythm game that trains on each player's beatboxed kick, snare and hi-hat, classifies live hits and scores their timing against a beatmap.
- Rebuilt the Python/OSC prototype as an in-engine C# system with onset detection, a 512-sample FFT, seven features and KNN classification.

Not Today, Darling! | Sole Audio Designer **2025**

Shipped game | Unreal Engine 5 | MetaSounds | Blueprints | REAPER | Eleven-week, six-person team

- Edited roughly 450 source audio files, built reactive kart engines, recorded five teammates and mixed the game under a 32-source limit.

Audio Tools | Solo Developer **2024 - 2026**

AudioQ | AutoTrim Exporter | Mówik

- Released AudioQ, a C++17/JUCE 8 Windows VST3 with a waveform loop editor, granular mode and DSP effects.
- Built AutoTrim Exporter, a Lua/ReaImGui REAPER tool released for Windows, macOS and Linux, to analyse processed renders, detect onset and tail, adjust level and export named audio assets.
- Released Mówik, a Python Windows app with local faster-whisper transcription, a bilingual interface and push-to-talk audio kept temporarily in RAM.

EDUCATION

Bachelor of Digital Arts and Entertainment **2023 - 2026**

Major in Game Sound Integration | Howest DAE, Belgium

Graduation thesis: Real-Time AI-Generated Sound Effects for Video Games: A Latency and Quality Analysis

Classical Piano and Music Theory **2013 - 2019**

State Music School | Kołobrzeg, Poland

LANGUAGES

Polish - Native | English - Fluent | German - Basic